Does RAG Work Through Q-Former? A Controlled Analysis for Radiology Report Generation

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Abstract

Retrieval-Augmented Generation (RAG) is widely adopted for radiology report generation, yet to our knowledge no prior work has systematically tested whether the Q-Former, a widely adopted vision-language bridging architecture, can leverage retrieved content. We investigate this using a dual cross-attention BLIP-2 Q-Former as a controlled testbed, evaluating three retrieval strategies, one oracle-query condition, and six decoder configurations across three hospital systems (MIMIC-CXR, IU X-Ray, CheXpert Plus). Under standard training, this Q-Former design does not meaningfully exploit retrieval content: RAG provides no practically meaningful improvement in BLEU-4 (Stage 1: 0.142 ± 0.001 *vs. RAG:* 0.144*, std* < 0.001*) or RadGraph F1 (*0.819 ± 0.001 *vs.* 0.818 ± 0.002*), whether the decoder is frozen, unfrozen, scaled to OPT-6.7B, or trained with aggressive learning rates, with consistent results on IU X-Ray and CheXpert Plus. Gradient measurements indicate gradient starvation: the visual pathway suffices, leaving little gradient on the retrieval cross-attention, which stays near initialization while the residual connection permits cost-free bypass; removing the residual collapses generation to 7–9 word outputs. Training with relevant retrieved reports from Stage 1 yields a small benefit (+1.4%* $\pm$ *1.0% relative across three seeds, versus +0.0%* $\pm$ *0.4% with random reports), and forced-utilization training (blank images with ground-truth context) shows the pathway works when retrieval is necessary: given another patient's report, outputs follow it in 99–100 of 100 cases across three seeds, versus 50 of 100 (chance) under standard training. These conclusions apply to the tested design, a Q-Former trained from scratch with ungated, residually wrapped retrieval cross-attention, and to our retrieval setup. We contribute length-controlled evaluation and retrieval content ablation as diagnostic tools for RAG evaluation.*

1. Introduction

Automated radiology report generation has advanced rapidly through vision-language architectures that produce clinically structured text from chest X-ray images [5, 6, 22]. Retrieval-Augmented Generation (RAG) has become a popular approach, conditioning generation on relevant reports retrieved from a training corpus [13, 33, 34, 42]. Existing RAG methods integrate retrieved text directly into the language model's input sequence. However, the Q-Former [20], a widely adopted vision-language bridging architecture used in BLIP-2 [20], InstructBLIP [9], MiniGPT-4 [46], and numerous downstream systems [7, 38], has not, to our knowledge, been systematically evaluated for its capacity to leverage retrieval context.

The Q-Former [20] is explicitly designed as an "information bottleneck": 32 learnable query embeddings cross-attend to visual features and pass only a compressed summary to the language model. This selectivity suggests a natural way to integrate retrieval. Existing radiology RAG methods place retrieved reports directly in the language model's context window [13, 33, 34], where findings from other patients reach the decoder without any check against the current image, and the input grows with the number of retrieved reports. Routing retrieval through the Q-Former instead offers a potential advantage: because the queries attend to the image before they attend to retrieved reports, retrieval cross-attention can be *image-conditioned*, weighting retrieved findings by their consistency with the current image. Radiology report generators already use this image-queried pattern with the same expectation [12, 15], and the Q-Former, as the standard bridge in BLIP-2-family models, is a natural place to implement it. The design also keeps the decoder input fixed at 32 query embeddings regardless of the number of retrieved reports, and the report corpus can be replaced, for example with another institution's archive, without retraining the language model. We find that under standard training this advantage is not realized: the retrieval pathway remains near initialization, and even with relevant retrieval content from Stage 1, the benefit is small and seed-

dependent.

Using a modified BLIP-2 Q-Former with dual cross-attention as a controlled testbed, we evaluate three retrieval strategies and one oracle-query condition across three hospital systems (MIMIC-CXR [18], IU X-Ray [11], CheXpert Plus [4, 16]) without institution-specific fine-tuning. Our contributions:

- Under standard training, our dual cross-attention Q-Former does not meaningfully exploit retrieval content. Across six decoder configurations (frozen and unfrozen OPT-2.7B, three retrieval training interventions, and frozen OPT-6.7B), RAG provides no practically meaningful improvement in BLEU-4 (Stage 1: 0.142 ± 0.001 vs. RAG: 0.144, std < 0.001) or RadGraph F1. The main MIMIC-CXR comparison (Stage 1 vs. BiomedCLIP) is confirmed across three seeds, with the same trend observed on IU X-Ray and corroborated on the limited CheXpert Plus evaluation (72 samples). Decoder ablations use a single seed.
- The failure is consistent with gradient starvation [30]: the visual pathway suffices, leaving little gradient pressure on the retrieval cross-attention, whose weights move only 0.003–0.005 from initialization while the residual connection permits cost-free bypass. Removing the residual degrades generation to 7–9 word outputs; training with relevant context yields a consistently positive but small benefit (+1.4% $\pm$ 1.0% relative, vs. +0.0% $\pm$ 0.4% with random context); and forced-utilization training (Exp D) shows the pathway works when retrieval is necessary: outputs follow another patient's report in 99–100 of 100 cases across three seeds, versus 50 of 100 under standard training.
- We contribute two evaluation tools, applicable in principle to any RAG system: a) length-controlled evaluation, which exposes output length as a confound in RAG comparisons, and b) retrieval content ablation, which tests whether models extract retrieval content or merely detect its presence.

2. Method

2.1. Architecture Overview

We extend BLIP-2's [20] vision-language framework with a second cross-attention layer for retrieved medical reports. The system consists of four components: (1) a frozen ViT-G/14 vision encoder, (2) a modified Q-Former with 32 learnable queries and dual cross-attention layers, (3) a frozen OPT-2.7B [44] language model decoder, and (4) a retrieval system that searches a FAISS [19] index built with S-PubMedBert-MS-MARCO [14] embeddings. Figure 1 illustrates the architecture.

2.2. Vision Encoder

We use the frozen ViT-G/14 vision encoder from EVA-CLIP [35] as our image feature extractor. Input images are resized to 224×224 pixels and processed through the vision transformer, producing patch-level features of dimension 1408. Patch features are mapped to the Q-Former's 768-d input by a fixed, randomly initialized linear projection (1408$\rightarrow$768) that is frozen throughout training and shared by all conditions.

2.3. Dual Cross-Attention Q-Former

The Q-Former [20] maintains 32 learnable query tokens of dimension 768, designed as an information bottleneck between the vision encoder and the language decoder. For single-source tasks, this compression filters irrelevant visual information. We extend the standard single cross-attention Q-Former with a second cross-attention layer to test whether the same queries can additionally carry retrieval context. We implement it with BLIP-2's configuration (12 layers, hidden size 768, 12 attention heads) using standard pre-normalized multi-head attention, with visual cross-attention in every layer. All Q-Former weights, including the query embeddings, are randomly initialized and trained in Stage 1; the ViT-G/14 encoder, the OPT decoder, and the Q-Former-to-OPT projection are taken from the pretrained BLIP-2 checkpoint.

Self-Attention. The 32 queries first attend to each other, enabling inter-query information exchange.

Visual Cross-Attention. The queries attend to the projected vision features via multi-head cross-attention with 12 heads.

Retrieval Cross-Attention. The queries, now enriched with visual information, attend to token-level representations of retrieved medical reports. Each of the $k = 3$ retrieved reports is encoded independently (up to 128 tokens) by a frozen PubMedBERT encoder [14] into 768-dimensional token embeddings, and the token sequences are concatenated to form the keys and values. Because the cross-attention has no positional encoding over keys, its output is invariant to the order of the retrieved reports; we therefore add no report-level segment embeddings. The retrieval encoder is not updated during training. We place retrieval cross-attention after visual cross-attention so that the queries already carry image information when they attend to retrieved reports; this ordering is what would allow image-conditioned selection of retrieved content. As with every other sublayer, the retrieval cross-attention and its feed-forward block are wrapped in residual connections, which preserves the visual pathway when retrieval is added. Each Q-Former layer computes: $\mathbf{H}_1 = \mathbf{Q} + \text{SelfAttn}(\text{LN}(\mathbf{Q}))$, $\mathbf{H}_2 = \mathbf{H}_1 + \text{CA}_v(\text{LN}(\mathbf{H}_1), \mathbf{V})$, $\mathbf{H}_3 = \mathbf{H}_2 + \text{CA}_r(\text{LN}(\mathbf{H}_2), \mathbf{R})$, $\mathbf{H}_4 = \mathbf{H}_3 + \text{FFN}_r(\text{LN}(\mathbf{H}_3))$, $\mathbf{H}_5 = \mathbf{H}_4 + \text{FFN}(\text{LN}(\mathbf{H}_4))$, where $\mathbf{Q}$ denotes the queries

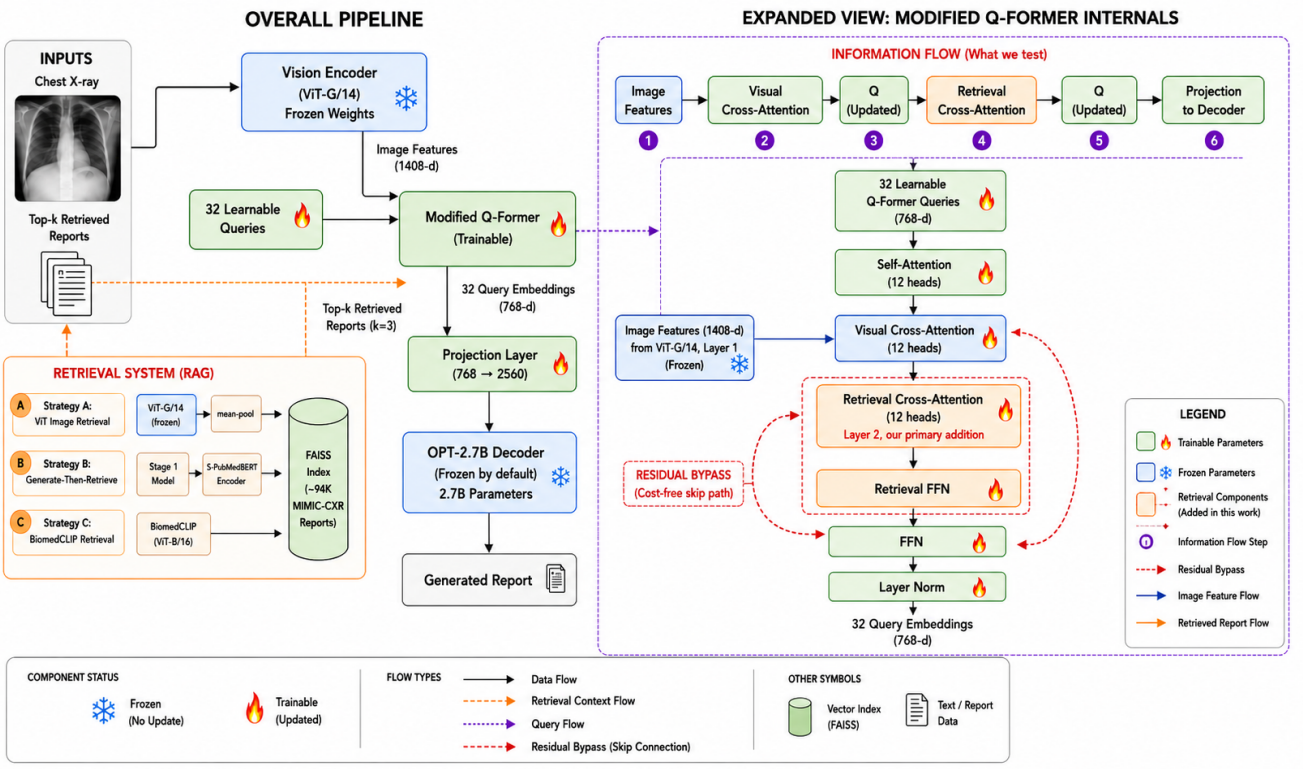


Figure 1. Architecture overview. **Left:** a chest X-ray is encoded by frozen ViT-G/14 and passed through our modified Q-Former (32 learnable queries) to OPT-2.7B. Three retrieval strategies (plus an oracle-query condition) draw reports from FAISS indices over ∼94K training reports: a text index (S-PubMedBERT) for generate-then-retrieve and oracle-query, and image indices (ViT projection or BiomedCLIP) for image-based retrieval. **Right:** Q-Former internals: visual cross-attention, then our added retrieval cross-attention (dashed red), whose residual connection lets the model bypass it under standard training (Section 3.6).

entering the layer, $\mathbf{V}$ the projected visual features, $\mathbf{R}$ the concatenated report token embeddings, CA_v and CA_r the visual and retrieval cross-attentions, FFN two-layer MLPs, and LN layer normalization; a final LN follows the last layer. Without retrieval, steps 3–4 are skipped. Exp A removes the additive residuals in steps 3–4. All information from both sources must pass through the same 32 query embeddings before reaching the decoder; our experiments test whether retrieval content reaches the decoder through this integration.

2.4. Language Decoder

The Q-Former's 32 output query embeddings are projected through a trainable linear layer, initialized from BLIP-2, into the input embedding space of OPT-2.7B [44], a pretrained autoregressive language model that serves as our frozen decoder. We select OPT-2.7B for three reasons: (1) it is the standard BLIP-2 decoder, enabling direct comparison; (2) most RAG-based radiology methods also use frozen decoders [13, 33, 41]; and (3) freezing isolates retrieval strategy as the sole variable. We also evaluate unfrozen OPT-2.7B and frozen OPT-6.7B (Section 2.9).

2.5. Retrieval System

Text Retrieval Encoder. For text-based retrieval, we encode reports and text queries using S-PubMedBert-MS-MARCO [14], a PubMedBERT model fine-tuned for semantic search, mapping radiology reports to 768-dimensional embeddings optimized for clinical semantic similarity. This encoder is used only to build the FAISS index and to encode text queries.

FAISS Index. We retain 94,229 training studies with non-empty findings sections and valid frontal images (one report per study). These are pre-encoded and stored in a FAISS [19] index configured for cosine similarity search. Retrieval operates at the study level.

BiomedCLIP Retrieval. We also use BiomedCLIP [45] (ViT-B/16) to produce 512-dimensional image embeddings trained on 15M biomedical image-text pairs. A separate FAISS index stores BiomedCLIP embeddings, enabling direct image-to-image retrieval where the corresponding ground-truth reports of the k nearest neighbors serve as retrieval context.

For all retrieval conditions, we use $k = 3$ nearest neigh-

bors.

2.6. Training Pipeline

Training proceeds in two stages, following BLIP-2 [20]:

Stage 1 (No RAG). The Q-Former and projection layers are trained for direct image-to-report generation with the retrieval cross-attention layer *disabled*. Only visual cross-attention is active. Stage 1 uses AdamW [27] with learning rate 1×10^{-4}, weight decay 0.01, batch size 8, and gradient clipping at 1.0, and runs for 5 epochs with cosine annealing [26] to 1×10^{-6}.

Stage 2 (RAG-Enhanced). The full dual cross-attention Q-Former is trained with retrieval context enabled. For each training sample, a pre-computed retrieval cache provides the top-$k = 3$ most similar training reports. Stage 2 uses AdamW with learning rate 5×10^{-5}, weight decay 0.01, batch size 8, gradient clipping at 1.0, and runs for 5 epochs with linear warmup over the first 10% of steps followed by cosine decay. The retrieval cache is built using Stage 1-generated draft reports as text queries. Drafts are generated for all 94,229 training samples (averaging 38.2 words per draft, compared to a ground-truth average of 49 words), ensuring the model trains with retrieval context of similar quality to what it will encounter at deployment. All reported metrics in this paper are from models trained on the complete, verified MIMIC-CXR corpus.

Retrieval excludes neither the same study nor the same patient; on 1,000 inspected samples, exact self-retrieval is rare (0.4% for text queries, 3.3% for BiomedCLIP) and 0.9% of retrievals come from the same patient.

2.7. Inference Retrieval Strategies

At inference, no ground-truth report text is available. We evaluate all strategies on MIMIC-CXR (11,822 validation samples) and perform zero-shot cross-institutional evaluation on IU X-Ray [11] (640 test samples) and CheXpert Plus [4, 16] (72 validation samples), using MIMIC-CXR training reports as the retrieval source throughout.

Strategy 1: ViT Image Retrieval. The input image is encoded by ViT-G/14, passed through the same fixed projection (1408→768) that feeds the Q-Former, and mean-pooled across patches. This 768-d embedding queries a separate FAISS index of identically encoded training images, returning the ground-truth reports of the k nearest neighbors.

Strategy 2: Generate-Then-Retrieve. A two-pass pipeline inspired by Generation-Augmented Retrieval [28]: (1) Stage 1 generates a draft report; (2) the draft is encoded via S-PubMedBert-MS-MARCO and queried against FAISS; (3) Stage 2 generates the final report using both the image and retrieved reports.

Strategy 3: BiomedCLIP Retrieval. The input image is encoded using BiomedCLIP [45] (ViT-B/16), querying a separate FAISS index for domain-aligned image-based retrieval.

Oracle-Query Condition. Ground-truth validation text serves as the retrieval query (not context), providing the upper bound of query quality. The retrieved reports are similar but not identical to the reference.

All strategies use identical generation parameters (beam search, 5 beams, `max_length`=100, `length_penalty`=1.0).

2.8. Controlled Evaluation Protocols

We report BLEU-4 and RadGraph F1 as primary metrics; ROUGE-L appears only in Table 1.

We introduce two evaluation protocols to isolate RAG's contribution:

Length-Controlled Evaluation. RAG-enhanced models may produce outputs of different lengths than non-retrieval baselines, confounding BLEU-4 through its sensitivity to output length (both n-gram precision and brevity penalty). For length-controlled evaluation, we truncate all predictions to fixed word count ceilings (35 and 39 words on MIMIC-CXR; 25, 31, and 38 on IU X-Ray) before recomputing both BLEU-4 and RadGraph F1 [17]. Stage 1 predictions are generated without length constraints (`max_new_tokens`=200; 68 ± 18 words) and RAG predictions with the Table 1 setting (∼39 words); truncating both to these thresholds therefore yields length-matched comparisons. We use simple truncation rather than abstractive summarization to avoid introducing additional variables that would confound the comparison.

Retrieval Content Ablation. We test whether a model uses retrieved content in two ways on 100 validation samples. First, with real images, we replace the $k = 3$ retrieved reports with irrelevant non-medical text (English pangrams, e.g., "The quick brown fox...", one per retrieval slot) and measure the exact-match identity rate and word-level overlap (Jaccard) between outputs. Second, in a swapped-context test, we pair a blank image with another patient's report as context and check whether the output overlaps that report more than the sample's own report. We term *presence detection* the behavior where outputs change when retrieval is provided but do not follow its content.

2.9. Decoder Ablation Configurations

We evaluate six configurations, all sharing the same Stage 1 checkpoint, retrieval cache, and evaluation protocol: (1) frozen OPT-2.7B (baseline); (2) unfrozen OPT-2.7B, fine-tuned in Stage 2 at a learning rate 50× lower than the Q-Former (∼1×10^{-6}) [20]; (3) a higher retrieval learning rate matching Stage 1 visual training (1×10^{-4}, a 10× boost); (4) retrieval cross-attention (query, key, value and output projections, and layer norm) pre-initialized from each layer's trained visual cross-attention; (5) both com-

bined; and (6) frozen OPT-6.7B (2.5× larger; bf16, batch size 4, learning rate 5×10^{-5}).

2.10. Diagnostic Experiments

We conduct five experiments to localize the failure. **Exp A** removes the additive residuals around the retrieval cross-attention and retrieval FFN (steps 3–4 above), forcing information through the retrieval pathway. **Exp B** enables retrieval cross-attention during Stage 1 at the visual learning rate (1×10^{-4}) with randomly sampled training reports as context, removing the initialization and training-order asymmetry; **Exp B′** uses BiomedCLIP-retrieved reports instead (both run across three seeds: 42, 123, 456). **Exp C** combines A and B. **Exp D** (forced utilization) makes two changes to Stage 2's data: the retrieval context is the sample's own ground-truth report (a copy task), and images are replaced with blank inputs to the frozen ViT ($p = 1.0$), so loss can only fall through the retrieval pathway. It is a capability probe, not a candidate system, run across three seeds (0, 123, 456). Exp D and the standard-training run used for gradient and weight measurements (Section 3.6) share one script (weight decay 0.05, cosine annealing to 1×10^{-7} without warmup) and differ only in retrieval context and image masking; we log retrieval gradient norms every 50 steps and weight statistics per epoch. All experiments use the retrained MIMIC-CXR corpus and consistent evaluation parameters.

3. Results

3.1. Main Results on MIMIC-CXR

Table 1 shows results on the MIMIC-CXR validation set (11,822 samples) with consistent generation parameters across all conditions. Multi-seed evaluation (3 seeds) confirms the finding is stable: Stage 1 achieves BLEU-4 of 0.142 ± 0.001 and BiomedCLIP retrieval achieves 0.144 (std < 0.001), a difference of 0.002. RadGraph F1 is also equivalent: Stage 1 0.819 ± 0.001 vs. RAG 0.818 ± 0.002. These comparisons establish the performance regime of our testbed and are not intended as claims of state-of-the-art performance.

RAG's effect is consistently positive in direction (∼+0.002 BLEU-4). Two-sided paired Wilcoxon signed-rank tests over 11,822 samples detect this difference ($p < 0.01$, uncorrected), but the effect size is negligible in that RadGraph F1 shows no corresponding gain (0.819 vs. 0.818), and the difference disappears under length control (Table 3). *We therefore characterize this as no practically meaningful improvement.*

BLEU-4 ranges from 0.143 to 0.144 across ViT, generate-then-retrieve, BiomedCLIP, and oracle conditions. Retrieved reports show non-zero lexical overlap with references (median Jaccard similarity: 0.188, IQR: 0.147–0.237). The oracle-query condition uses ground-truth text as the retrieval query yet achieves BLEU-4 of 0.144, matching the baseline. Its null result suggests retrieval quality alone does not explain the lack of improvement. Exp D (Section 3.5) separately tests true oracle *context*, where the reference report is fed directly to the retrieval pathway.

Table 1. Results on MIMIC-CXR (11,822 validation samples). ‡ Published results included for contextual reference; evaluation splits, report sections, and preprocessing differ from ours and are not directly comparable. † VLM baselines (1,000-sample subset). Our methods report mean ± std across 3 seeds where available.

Method	B-4	R-L	RG-F1
Published methods‡			
R2Gen [5]	0.103	0.277	—
R2GenCMN [6]	0.106	0.278	—
METransformer [40]	0.124	0.291	—
R2GenGPT [41]	0.134	0.297	—
KARGEN [22]	0.140	0.305	—
General-purpose VLMs†			
Qwen2-VL (zero-shot)	0.029	0.180	0.796
Qwen2-VL (prompted)	0.103	0.259	0.825
Ours (3 seeds where noted, otherwise single seed)			
Stage 1 (no RAG)	$0.142_{\pm.001}$	0.294	$0.819_{\pm.001}$
+ ViT retrieval	0.143	0.294	0.821
+ Gen-then-retrieve	0.143	0.294	0.821
+ BiomedCLIP	$0.144_{\pm<.001}$	0.296	$0.818_{\pm.002}$
+ Oracle query (GT text)	0.144	0.295	0.821

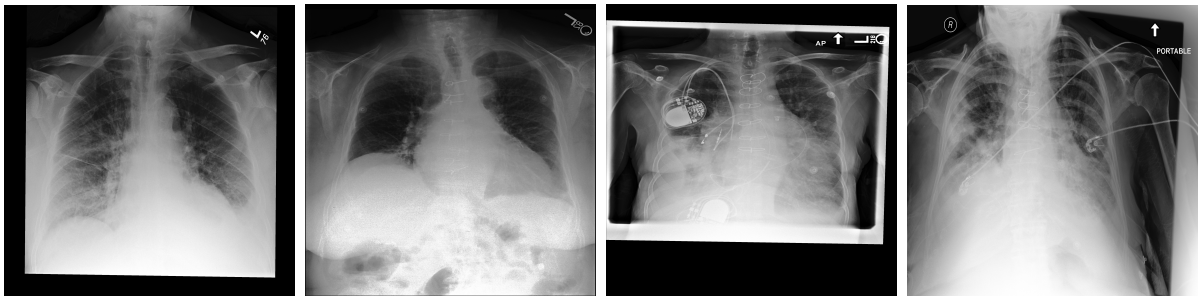
Table 2. Cross-institutional zero-shot evaluation. Models trained on MIMIC-CXR only, evaluated without fine-tuning. No values are bolded given the small sample sizes and negligible differences. All methods use consistent generation parameters (`max_length`=100). Generate-then-retrieve is omitted: its query inherits source-institution reporting conventions, introducing a distribution shift absent from image-based retrieval.

	IU X-Ray (640)		CheXpert+ (72)	
Method	**B-4**	**RG-F1**	**B-4**	**RG-F1**
Stage 1 (no RAG)	0.101	0.854	0.071	0.788
+ ViT retrieval	0.093	0.857	0.072	0.776
+ BiomedCLIP	0.092	0.857	0.066	0.784

3.2. Cross-Institutional Generalization

We test all methods on IU X-Ray (640 test samples, Indiana University) and CheXpert Plus (72 validation samples, Stanford) in a zero-shot transfer setting (Table 2).

On IU X-Ray, Stage 1 achieves the highest BLEU-4 (0.101) while retrieval methods score lower (ViT 0.093, BiomedCLIP 0.092), consistent with the MIMIC-CXR finding. RadGraph F1 is nearly identical across methods (0.854



(a) RAG helps	(b) RAG neutral	(c) RAG hallucinates	(d) Qwen2-VL templated
GT: Low lung volumes. Left costophrenic angle obscured: moderate **pleural effusion**. Retrocardiac consolidation likely atelectasis. Moderate **pulmonary edema** unchanged. Moderate **cardiomegaly**. No pneumothorax.	**GT:** Opacity at left base, probably **atelectasis** and scarring. Pulmonary vasculature indistinct: **pulmonary vascular congestion**. Cardiac enlargement stable. Median **sternotomy wires** intact.	**GT:** Right-sided **pacemaker** with 2 leads, one in right atrium, one abandoned. Second device over right upper quadrant. Status post median **sternotomy**. **Pulmonary edema**. **Bilateral pleural effusions**. **Atelectasis**. Cardiac silhouette **enlarged**.	**GT:** Status post median **sternotomy** and **CABG**. Heart **mildly enlarged**. Mild **pulmonary edema**, slightly worse. Persistent **opacification** left base. Moderate **bilateral pleural effusions**. **Coronary stent** noted.
Stage 1: Lung volumes low. Mild interstitial **edema**. **No large effusion**. Cardiomediastinal silhouette **within normal limits**. Bony structures intact.	**Stage 1:** Midline **sternotomy wires** and mediastinal clips noted. No focal consolidation, **pleural effusion**, or pneumothorax. Cardiomediastinal silhouette stable. Bony structures intact.	**Stage 1:** Left chest wall dual-lead **pacemaker** with leads in right atrium and right ventricle. Median **sternotomy** wires and clips. No pneumothorax. Cardiomediastinal silhouette stable. **No effusion noted**.	**Stage 1:** Status post median **sternotomy** and **CABG**. Moderate **cardiomegaly** unchanged. Mild **pulmonary edema**. Small **bilateral pleural effusions**. No pneumothorax. Bony structures intact.
BiomedCLIP: Mild **cardiomegaly**. Mediastinal and hilar contours unremarkable. Small **bilateral pleural effusions** noted. No pneumothorax. Bony structures intact.	**BiomedCLIP:** Midline **sternotomy wires** and mediastinal clips noted. Lungs clear without focal consolidation, **effusion**, or pneumothorax. Cardiomediastinal silhouette stable. Bony structures intact.	**BiomedCLIP:** Dual lead **pacemaker**/ICD with leads in right atrium and right ventricle. Status post median **sternotomy** and **CABG**. Moderate **cardiomegaly** unchanged. Mediastinal contours…	**BiomedCLIP:** Status post median **sternotomy** and **CABG**. Moderate to severe **cardiomegaly** unchanged. **Bilateral pleural effusions** with bibasilar **atelectasis**. No pneumothorax.
Qwen2-VL: Heart size **normal**. Lungs clear, **no pleural effusion** or pneumothorax. Diaphragms intact. Costophrenic angles **sharp**. Ribs and clavicles normal.	**Qwen2-VL:** Heart size **normal**. Lungs clear, no visible opacities. **No signs of consolidation, pleural effusion**, or pneumothorax. Ribs and clavicles well-defined. **No foreign objects**.	**Qwen2-VL:** **Pacemaker** implant in left lower chest. Leads properly positioned. Heart silhouette **normal, no cardiomegaly**. Lungs **clear**. **No pleural effusion** or pneumothorax.	**Qwen2-VL:** Right-sided portable view. **No obvious fractures**. Lung fields **appear clear**. Heart silhouette **within normal limits, no cardiomegaly**. Diaphragm and costophrenic angles **clear**.

Figure 2. Qualitative comparison on four pathological MIMIC-CXR cases. **Green**: correct findings; **red**: incorrect or missed findings. **(a)** The RAG model (BiomedCLIP retrieval) captures cardiomegaly and bilateral pleural effusions that Stage 1 misses; given the swapped-context results (Section 3.3), such case-level differences more likely reflect the Stage 2 model and retrieval-induced perturbation than retrieved content. **(b)** Both methods produce nearly identical reports, correctly identifying sternotomy wires but missing atelectasis and congestion, consistent with presence detection rather than content extraction (Section 3.3). **(c)** The RAG model generates "CABG" and "cardiomegaly" not present in the ground truth; single cases cannot show whether such findings come from the retrieved reports. Together, panels (a) and (c) illustrate how individual-case gains and losses can average to the equivalent aggregate performance observed in Table 1. **(d)** Both our methods capture key findings; Qwen2-VL produces a templated "normal" report despite visible sternotomy, cardiomegaly, effusions, and edema, illustrating how templated outputs can achieve high RadGraph F1 without case-specific accuracy (Section 3.3).

vs. 0.857, $\Delta = 0.003$), though this small gap widens slightly under length control (Section 3.3). On CheXpert Plus, all methods produce equivalent results (BLEU-4 $\sim$0.07); given the limited sample size (72), we treat this as a corroborating trend rather than independent confirmation. Generate-then-retrieve cannot transfer zero-shot because its query inherits source-institution reporting conventions, a deployment limitation of text-query RAG pipelines.

3.3. Understanding RAG's Mechanism

Having established that RAG provides no practically meaningful gain, we next ask whether this equivalence is genuine or a measurement artifact.

Length-Controlled Evaluation. Under the constrained decoding for Table 1 (`max_length`=100), all methods produce outputs of 37–39 words on MIMIC-CXR. For length-controlled analysis, we truncate all predictions to fixed word counts (Section 2.8) to verify that the metric equivalence in Table 1 reflects genuine equivalence rather than a length artifact (Table 3).

Table 3. Length-controlled evaluation on MIMIC-CXR (predictions truncated to 35 and 39 words). Generate-then-retrieve is omitted; its BLEU-4 is identical to ViT retrieval in Table 1.

	BLEU-4		RG-F1
Method	**35w**	**39w**	**39w**
Stage 1 (no RAG)	0.133	0.143	0.820
+ ViT retrieval	0.134	0.141	0.821
+ BiomedCLIP	0.134	0.141	0.821
+ Oracle query	0.134	0.142	0.821

All methods remain numerically close at matched

lengths. RadGraph F1 at 39 words confirms equivalence (Table 3). Length-controlled evaluation on IU X-Ray confirms the same pattern: at 31 words (closest to IU's ground-truth average), Stage 1 achieves BLEU-4 of 0.115 versus 0.108 (ViT) and 0.107 (BiomedCLIP). Stage 1 leads at every threshold tested (25w, 31w, 38w). On IU X-Ray RadGraph F1, the gap widens slightly under length control (0.003 $\rightarrow$ 0.008 at 38 words), the largest clinical-metric advantage for retrieval in our results, though Stage 1 leads BLEU-4 at every IU threshold.

Retrieval Content Ablation. With real images, replacing the retrieved reports with pangrams changes most outputs (85 of 100 differ; median word overlap 0.57), so retrieval perturbs generation. The swapped-context test shows the perturbation carries no content: given another patient's report and a blank image, the standard model's outputs overlap that report no more than an unrelated one (0.18 vs. 0.18) and follow it in 50 of 100 cases, chance level, whereas forced-utilization models follow it in 99–100 of 100 (Section 3.5). Neither extreme realizes the motivating advantage: an image-conditioned model would adopt retrieved findings consistent with the image and reject those that contradict it, whereas the standard model ignores retrieved content and the forced model copies it regardless. We note that word-level similarity can miss clinically meaningful differences (e.g., "no pneumothorax" vs. "small pneumothorax"); entity-level analysis using RadGraph or CheXbert labels would provide a more clinically grounded assessment. This presence-detection behavior is distinct from findings that retrieval relevance shapes RAG output quality [8]: in our setting, outputs do not follow retrieved content, relevant or not.

Metric limitations. Prompted Qwen2-VL, included to illustrate metric behavior rather than as a radiology system, reaches the highest RadGraph F1 (0.825) with highly templated outputs (36% unique), so entity-overlap scores can reward templates. Our Stage 1 and RAG models produce diverse outputs, though subtle clinical differences detectable only by radiologists cannot be ruled out.

3.4. Decoder and Q-Former Ablation

The equivalence across retrieval strategies and the content ablation results suggest retrieval information has little measurable influence on the decoder. We evaluate six decoder configurations (Table 4).

Unfreezing OPT-2.7B (Row 2) reduces validation loss but degrades BLEU-4 (0.132), suggesting decoder fine-tuning shifts the model away from its pretrained generation behavior. Rows 3–5 test retrieval-specific interventions: higher learning rate (10$\times$), pre-initialization from visual weights, and both combined. None produces a positive RAG effect. Under the gradient-starvation account, these nulls are expected: boosting near-zero gradients still yields near-zero, and pre-initialization provides a starting point but no task-relevant signal. Correctly initialized retrieval layers raise the no-retrieval score (0.147–0.148 vs. 0.144) without making the model follow retrieved content (44–46 of 100 in the swapped-context test). OPT-6.7B (Row 6) shows lower absolute metrics under a reduced memory budget (batch size 4, bf16, val loss 1.52 vs. 1.38 for 2.7B). This run is not performance-matched and serves only to test whether RAG's null effect holds at larger scale; it does.

Across the tested single-seed configurations, we find no evidence that decoder freezing, retrieval learning rate, initialization, or decoder scale explains the null RAG effect. Their BLEU-4 differences ($|\Delta| \leq 0.002$) fall within the seed-to-seed spread of Table 1 and Exp B/B′.

Table 4. Decoder and Q-Former ablation on MIMIC-CXR (11,822 samples, single seed; baseline differs slightly from Table 1 multi-seed average). No configuration produces a positive RAG effect; decoder ablations are single-seed. Δ denotes absolute BLEU-4 difference.

	BLEU-4		RG-F1	
Configuration	**No RAG**	**+RAG (Δ)**	**No RAG**	**+RAG (Δ)**
OPT-2.7B frozen	0.144	0.143 ($-.001$)	0.819	0.818 ($-.001$)
OPT-2.7B unfrozen	0.132	0.132 (.000)	0.811	0.809 ($-.002$)
+ Higher retr. LR	0.145	0.144 ($-.001$)	0.818	0.819 (+.001)
+ Pre-init retr.	0.148	0.146 ($-.002$)	0.817	0.821 (+.004)
+ Both combined	0.147	0.146 ($-.001$)	0.817	0.819 (+.003)
OPT-6.7B frozen	0.130	0.128 ($-.002$)	0.763	0.762 ($-.001$)

Table 5. Diagnostic experiments on MIMIC-CXR (11,822 samples; baseline differs slightly from Table 1 multi-seed average). Δ denotes absolute BLEU-4 difference.

Experiment	**Image**	**Seeds**	**No RAG**	**+RAG**	**Δ**	**Val Loss**
Baseline	real	1	0.144	0.143	$-.001$	1.38
A: Remove residual	real	1	0.141	$<0.001^{*}$	—	5.38
B: Random retrieval	real	3	$0.145_{\pm.001}$	$0.145_{\pm.001}$	.000	1.39
B′: Relevant retrieval	real	3	$0.143_{\pm.001}$	$0.145_{\pm.003}$	+.002	1.39
C: A + B combined	real	1	0.136	$<0.001^{*}$	—	5.12
D: Forced util. †	blank	3	n/a	n/a	n/a	**0.27**

* Generation degrades to 7–9 word outputs.

† Own GT report as context; follows a swapped report in 99–100/100; gradient ratio 5.4–5.5$\times$ (all 3 seeds).

3.5. Diagnostic Experiments

To resolve and localize the failure mechanism, we run the following five experiments.

Exp A: Remove residual. Without the residual, enabling retrieval degrades generation to 7-word outputs (val loss 5.38). With retrieval disabled, BLEU-4 remains 0.141. The residual enables bypass but also protects the decoder from the untrained pathway's corrupted output.

Exp B: Random retrieval in Stage 1. Retrieval enabled from Stage 1 at the visual learning rate (1×10^{-4}) with random context. Across three seeds, RAG Δ = +0.0% $\pm$ 0.4% (range -0.4% to +0.4%). This shows that simply enabling the pathway from Stage 1 is insufficient; random context

provides no gradient pressure to use retrieval.

Exp B′: Relevant retrieval in Stage 1. To test whether informative context enables the pathway to train, we repeat Exp B using BiomedCLIP-retrieved reports (clinically similar to each training image; 3.3% of top-1 retrievals are exact self-matches, as the query image is not excluded from the index). Across three seeds, RAG Δ = +1.4% $\pm$ 1.0% (range: +0.2% to +2.7%). The effect is consistently positive across all three seeds but small relative to seed variance; the 3.3% self-match rate is a possible contributor that we cannot fully exclude without re-running with self-exclusion.

Exp C: A + B combined. Residual removal with Stage 1 retrieval still degrades generation to 9-word outputs (val loss 5.12).

Exp D: Forced utilization. To test whether the retrieval pathway is architecturally capable of routing text, we train Stage 2 with ground-truth reports as retrieval context and masked images ($p = 1.0$), creating a copy task where the only route to low loss is through the retrieval cross-attention. Validation loss drops to 0.27 in each of three seeds (vs. 1.38 baseline), and retrieval gradient norms grow to 5.4–5.5$\times$ the visual pathway by training end (from 0.3–0.5$\times$ at initialization). Swapped-context testing confirms genuine information routing: outputs follow the provided report in 99–100 of 100 cases across seeds (word overlap 0.63–0.66 with the context vs. 0.14 with the image's own report), against 50 of 100 for the standard model under the identical test, and replacing the context with pangrams changes outputs almost entirely (median overlap 0.04–0.06 vs. 0.35), demonstrating that the pathway can encode and route retrieval content when retrieval is made necessary.

Together, Exps A–D show the failure is not caused by initialization or training order, the residual is load-bearing, and the pathway is capable when forced. We discuss the mechanism in Section 5.

3.6. Gradient and Weight Analysis

Direct gradient measurement separates the two training conditions. We define the retrieval-to-visual gradient ratio as $\|\nabla_{\theta_R}\mathcal{L}\|_2 \,/\, \|\nabla_{\theta_V}\mathcal{L}\|_2$, where θ_R and θ_V denote the trainable retrieval and visual cross-attention parameters, respectively (the ViT encoder is frozen and excluded). Under standard training, this ratio declines from 0.41 at initialization to 0.20 by training end; under forced utilization (Exp D), the same ratio grows from 0.3–0.5 at initialization to 5.4–5.5 across three seeds. Both start from comparable ratios but diverge in opposite directions: the same pathway receives growing gradient when the task requires it and shrinking gradient when the visual pathway already suffices.

Weight statistics provide supporting evidence. Measurements come from a dedicated standard-training run with logging (final validation loss 1.38, matching the main Stage 2 model). Under standard training, the retrieval projections move only 0.003–0.005 from initialization (mean absolute change over five epochs) and keep their initialization scale (`in_proj` std 0.025, `out_proj` std 0.020), whereas the same layers move 0.009–0.030 under forced utilization. In contrast, the Q-Former's self-attention and feed-forward layers show meaningful updates from Stage 1 (mean absolute change 0.02–0.04).

With retrieval weights near initialization, the residual branch perturbs the queries without carrying report content (Section 3.3), so all retrieval strategies, including oracle-query, perform alike.

4. Related Work

4.1. Radiology Report Generation

Report generation has progressed from memory-driven transformers (R2Gen [5], R2GenCMN [6]) and knowledge graphs (DCL [21], KARGEN [22]) to frozen-LLM decoders (R2GenGPT [41]), region-guided detection (RGRG [36]), and multi-view and longitudinal contrastive learning (MCL [24], MLRG [25]). The Q-Former [20], an "information bottleneck" of learnable queries, has become the dominant vision-language bridge (BLIP-2, InstructBLIP [9], MiniGPT-4 [46]) and works well for single-source compression, including medical anomaly detection [10]. InstructBLIP feeds brief instructions into the Q-Former, which differs from dense, multi-sentence retrieved reports; to our knowledge, retrieval integration through the Q-Former has not been systematically evaluated.

4.2. RAG in Medical Imaging

RAG methods differ in their *integration point*: direct LLM context, added cross-attention layers, or bottleneck integration through fixed-capacity queries; we evaluate the third. Most methods use direct LLM integration: CXR-RePaiR [13] retrieves with contrastive image-text models, LaB-RAG [33] filters retrieved reports with predicted labels, FactMM-RAG [34] grounds generation in verified facts, and CEMRAG [31] combines interpretable concepts with retrieval on MIMIC-CXR and IU X-Ray. Bottleneck integration instead compresses retrieval into queries that already carry visual information; information-bottleneck views of RAG [47] and image-queried retrieval attention [12] share the expectation that such compression keeps content consistent with the input. We find this selectivity does not emerge in the tested design. To our knowledge, no prior work controls for output length when comparing RAG and non-RAG methods, or tests whether models extract retrieval content versus responding to its presence; we address both gaps.

4.3. Foundation Models and Evaluation

Medical foundation models such as Flamingo-CXR [37], whose reports matched clinician quality in 56–77% of cases, and MAIRA-2 [2], which advanced grounded generation with prior-exam context, target radiology directly; we use BiomedCLIP [45] as a retrieval encoder and Qwen2-VL [39] as a general-purpose baseline. Lexical metrics (BLEU [29], ROUGE [23]) may miss clinical correctness, motivating RadGraph [17], CheXbert [32], and RadCliQ [43]. We extend these practices in two directions: length-controlled evaluation exposes output length as a confound (shorter outputs gain n-gram precision but risk the brevity penalty), and retrieval content ablation tests whether models extract retrieval content or merely detect its presence.

5. Discussion and Future Work

Failure mechanism. Our results point to an interaction between an architectural choice (a residually-wrapped cross-attention pathway added after the visual pathway) and a training condition (the visual pathway already suffices, so no loss pressure routes through retrieval). Training retrieval from Stage 1 (Exp B) or initializing it from visual weights (Table 4) does not change this, and removing the residual connection (Exp A/C) lets the untrained pathway corrupt the queries. When retrieval is made necessary (Exp D), the same pathway trains and routes content, and the retrieval-to-visual gradient ratio grows where it otherwise halves (Section 3.6), so the failure reflects gradient starvation [30] rather than architectural incapacity. We cannot fully exclude contributing explanations such as retrieval-distribution mismatch, representation asymmetry, or retrieval redundancy. Flamingo [1] engineered around precisely this failure mode for cross-attention in frozen language models.

Why the bypass matters. That an optimizer bypasses a branch the task does not need is, in hindsight, an expected outcome. What makes it consequential is that standard evaluation does not reveal it: retrieval changes most outputs (85 of 100 differ when retrieved reports are replaced by pangrams) and yields a statistically detectable BLEU-4 gain ($p < 0.01$, uncorrected), which a conventional comparison could report as an improvement. Only length control and content tests show that the model's outputs do not follow the retrieved content. Han et al. observe a related effect, where adding image-queried retrieval slightly lowered scores, which they attribute to redundant or erroneous retrieved reports given image features that already carry sufficient information [15].

Reconciling with prior work. Prior RAG methods [13, 33, 34] report improvements but place retrieval in the decoder's input, where content arrives by construction. Q-Former integration must earn delivery through gradient flow, and under standard training it does not. Reported gains may also depend on retrieval that is more complementary to the image and on comparisons without length control.

Practical implications. For practitioners adding retrieval to Q-Former-based systems, our results suggest three checks. First, a small metric gain after adding retrieval is not evidence that retrieved content is used; a swapped-context or content ablation test distinguishes content use from presence detection at little cost. Second, RAG comparisons should control output length, since BLEU-4 is sensitive to it. Third, when the image already supports the task, an added retrieval branch needs an explicit training incentive, such as gated cross-attention or intermittent image masking, rather than standard end-to-end training.

Limitations and future directions. Our conclusions are specific to the tested design: an ungated retrieval cross-attention placed after visual cross-attention in each layer of a Q-Former trained from scratch, wrapped in additive residuals, and trained with the recipes in Sections 2.6 and 2.10. We do not test pretrained BLIP-2 Q-Former weights, gated cross-attention, other placements of the retrieval layer, or domain-adapted decoders such as BioMedLM [3]. CheXpert Plus has only 72 samples, all datasets are chest radiographs, and a radiologist reader study would strengthen clinical validation. Stage 2 trains with text-based retrieval, so the image-based strategies face a distribution mismatch, although the matched conditions (generate-then-retrieve, oracle-query) also show no improvement. The gradient-starvation account points to two remedies: zero-initialized tanh-gated cross-attention following Flamingo [1], and partial forced-utilization training (image masking at $p < 1.0$) bridging our Exp D probe and deployable training; whether the pathway trains under partial masking, or with imperfect retrieval at inference, remains open. Matched comparisons with direct-LLM RAG systems [33, 34], using our evaluation tools, would test whether they show similar dynamics.

Summary. Under standard training, the tested Q-Former design does not meaningfully exploit retrieval content; forced-utilization training shows this reflects gradient starvation rather than architectural incapacity.

Ethics

This study uses MIMIC-CXR, IU X-Ray, and CheXpert Plus under their respective data use agreements; all data is de-identified.

Code. Code for reproducing all experiments is available at https://github.com/saadur1113/Radiology-report-analysis.

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MIMIC-CXR was accessed through PhysioNet under its credentialed data use agreement.

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